

PRACTICAL EXAMPLES

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1. CREATING THE TABLE AND INSERTING THE VALUES

The screenshot displays the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' and 'INSTANCE' sections. The main window shows a query window with the following SQL query:

```
SELECT *FROM orders;
```

The query results are displayed in a table grid with the following columns: `ord_no`, `purch_amt`, `ord_date`, `customer_id`, and `salesman_id`. The results show 12 rows of data.

ord_no	purch_amt	ord_date	customer_id	salesman_id
70001	150.50	2012-10-05	3005	5002
70002	65.26	2012-10-05	3002	5001
70003	2480.40	2012-10-10	3009	5003
70004	110.50	2012-08-17	3009	5003
70005	2400.60	2012-07-27	3007	5001
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
70010	1983.43	2012-10-10	3004	5006
70011	75.29	2012-08-17	3003	5007

The bottom section of the interface shows the 'Output' window with the following action output:

#	Time	Action	Message
48	15:24:29	SELECT branch_name, COUNT(*) AS Total_...	1 row(s) returned
49	15:25:34	SELECT branch_name, AVG(amount) AS Av...	3 row(s) returned
50	15:26:42	SELECT branch_name, SUM(amount) AS To...	3 row(s) returned
51	15:28:06	SELECT branch_name, AVG(amount) AS Av...	3 row(s) returned
52	15:36:36	INSERT INTO orders VALUES (70001,150.50,20...	12 row(s) affected Records: 12 Duplicates
53	15:37:04	SELECT *FROM orders LIMIT 0, 1000	12 row(s) returned

2.ALL ORDERS HANDLED

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 5* SQL File 6* x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

SQL File 6*

```
123
124 • SELECT *
125 FROM orders
126 WHERE ord_date='2012-09-10';
127 • SELECT *
128 FROM orders
129 WHERE salesman_id=5001;
130
131
```

Limit to 1000 rows

Result Grid

	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70002	65.26	2012-10-05	3002	5001
	70005	2400.60	2012-07-27	3007	5001
	70008	5760.00	2012-09-10	3002	5001
	70013	3045.60	2012-04-25	3002	5001
*	NULL	NULL	NULL	NULL	NULL

orders 23 x

Output

Action Output

#	Time	Action	Message
✓ 51	15:28:06	SELECT branch_name, AVG(amount) AS Av...	3 row(s) returned
✓ 52	15:36:36	INSERT INTO orders VALUES (70001,150.50,'20...	12 row(s) affected Reco
✓ 53	15:37:04	SELECT *FROM orders LIMIT 0, 1000	12 row(s) returned
✗ 54	15:41:09	FROM orders SELECT * WHERE purch_amt > 20...	Error Code: 1064. You H
✓ 55	15:41:50	SELECT * FROM orders WHERE ord_date='2012...	3 row(s) returned
✓ 56	15:42:36	SELECT * FROM orders WHERE salesman_id=5...	4 row(s) returned

3.ORDERS RANKED IN ASCENDING ORDER

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'Navigator' pane with sections for 'MANAGEMENT' (Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, Data Import/Restore), 'INSTANCE' (Startup / Shutdown, Server Logs, Options File), and 'PERFORMANCE' (Dashboard, Performance Reports, Performance Schema Setup). Below this are 'Administration' and 'Schemas' tabs. The main editor area shows a SQL query in 'SQL File 6'. The query is as follows:

```
126 WHERE ord_date='2012-09-10';
127 SELECT *
128 FROM orders
129 WHERE salesman_id=5001;
130
131 SELECT *
132 FROM orders
133 ORDER BY purch_amt DESC;
134
```

The 'Result Grid' is displayed below the query, showing 24 rows of data. The columns are: ord_no, purch_amt, ord_date, customer_id, and salesman_id. The data is sorted by purch_amt in descending order.

ord_no	purch_amt	ord_date	customer_id	salesman_id
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70010	1983.43	2012-10-10	3004	5006
70007	948.50	2012-09-10	3005	5002
70009	270.65	2012-09-10	3001	5005
70012	250.45	2012-06-27	3008	5002
70001	150.50	2012-10-05	3005	5002
70004	110.50	2012-08-17	3009	5003

The 'Output' pane at the bottom shows the execution log. It includes the following entries:

#	Time	Action	Message	Duration / P
52	15:36:36	INSERT INTO orders VALUES (70001,150.50,'20...	12 row(s) affected Records: 12 Duplicates: 0 Wa...	0.063 sec
53	15:37:04	SELECT *FROM orders LIMIT 0, 1000	12 row(s) returned	0.000 sec /
54	15:41:09	FROM orders SELECT * WHERE purch_amt > 20...	Error Code: 1064. You have an error in your SQL s...	0.031 sec
55	15:41:50	SELECT * FROM orders WHERE ord_date='2012...	3 row(s) returned	0.000 sec /
56	15:42:36	SELECT * FROM orders WHERE salesman_id=5...	4 row(s) returned	0.000 sec /
57	15:43:20	SELECT * FROM orders ORDER BY purch_amt	12 row(s) returned	0.000 sec

4.TO FIND OLDEST 1ST ORDERS

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 5* SQL File 6* x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information: orders 25 x

No object selected

Result Grid

ord_no	purch_amt	ord_date	customer_id	salesman_id
70013	3045.60	2012-04-25	3002	5001
70012	250.45	2012-06-27	3008	5002
70005	2400.60	2012-07-27	3007	5001
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
70001	150.50	2012-10-05	3005	5002
70002	65.26	2012-10-05	3002	5001

Output: Action Output

#	Time	Action	Message
53	15:37:04	SELECT *FROM orders LIMIT 0, 1000	12 row(s) returned
54	15:41:09	FROM orders SELECT * WHERE purch_amt > 20...	Error Code: 1064. You have an e
55	15:41:50	SELECT * FROM orders WHERE ord_date=2012...	3 row(s) returned
56	15:42:36	SELECT * FROM orders WHERE salesman_id=5...	4 row(s) returned
57	15:43:30	SELECT * FROM orders ORDER BY purch_amt ...	12 row(s) returned
58	15:44:22	SELECT * FROM orders ORDER BY ord_date LI...	12 row(s) returned

5.SUM OF ALL REVENUE GRANTED

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. Below this is the 'INSTANCE' section with Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes Dashboard, Performance Reports, and Performance Schema Setup. The bottom left shows 'Administration' and 'Schemas' tabs, with 'No object selected' displayed.

The main editor area shows a SQL query in 'SQL File 6*':

```
134
135 • SELECT *
136 FROM orders
137 ORDER BY ord_date;
138
139 • SELECT SUM(purch_amt) AS total_revenue
140 FROM orders;
141
142
```

The 'Result Grid' shows the results of the second query:

total_revenue
17541.18

The 'Output' tab at the bottom shows a table of execution results:

#	Time	Action	Message	Duration
54	15:41:09	FROM orders SELECT * WHERE purch_amt > 20...	Error Code: 1064. You have an error in your SQL s...	0.031 sec
55	15:41:50	SELECT * FROM orders WHERE ord_date='2012...	3 row(s) returned	0.000 sec
56	15:42:36	SELECT * FROM orders WHERE salesman_id=5...	4 row(s) returned	0.000 sec
57	15:43:30	SELECT * FROM orders ORDER BY purch_amt ...	12 row(s) returned	0.000 sec
58	15:44:22	SELECT * FROM orders ORDER BY ord_date LI...	12 row(s) returned	0.000 sec
59	15:45:02	SELECT SUM(purch_amt) AS total_revenue FRO...	1 row(s) returned	0.000 sec

6.FINDING AVG ORDER VALUE

The screenshot shows the MySQL Workbench interface. The left sidebar contains the Navigator pane with sections for MANAGEMENT, INSTANCE, and PERFORMANCE. The central editor shows a SQL query in a file named 'SQL File 6'. The query is as follows:

```
135 • SELECT *
136 FROM orders
137 ORDER BY ord_date;
138
139 • SELECT SUM(purch_amt) AS total_revenue
140 FROM orders;
141 • SELECT AVG(purch_amt) AS average_order
142 FROM orders;
143
```

The query results are displayed in the 'Result Grid' pane, showing a single row for the average order value:

average_order
1461.765000

The bottom pane shows the 'Output' section with a table of action output:

#	Time	Action	Message
55	15:41:50	SELECT * FROM orders WHERE ord_date='2012...	3 row(s) returned
56	15:42:36	SELECT * FROM orders WHERE salesman_id=5...	4 row(s) returned
57	15:43:30	SELECT * FROM orders ORDER BY purch_amt ...	12 row(s) returned
58	15:44:22	SELECT * FROM orders ORDER BY ord_date LI...	12 row(s) returned
59	15:45:02	SELECT SUM(purch_amt) AS total_revenue FRO...	1 row(s) returned
60	15:45:41	SELECT AVG(purch_amt) AS average_order FRO...	1 row(s) returned

7.HIGHEST ORDER AMOUNT

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' and 'INSTANCE' sections. The 'MANAGEMENT' section includes 'Server Status', 'Client Connections', 'Users and Privileges', 'Status and System Variables', 'Data Export', and 'Data Import/Restore'. The 'INSTANCE' section includes 'Startup / Shutdown', 'Server Logs', and 'Options File'. The 'PERFORMANCE' section includes 'Dashboard', 'Performance Reports', and 'Performance Schema Setup'. The 'Administration' tab is selected, showing 'Schemas' and 'Information'. The 'Information' section displays 'No object selected'.

The main editor window shows a SQL query in 'SQL File 6':

```
140 FROM orders;  
141 • SELECT AVG(purch_amt) AS average_order  
142 FROM orders;  
143  
144  
145 • SELECT MAX(purch_amt) AS highest_order  
146 FROM orders;  
147  
148
```

The query results are displayed in the 'Result Grid' below the editor:

highest_order
5760.00

The 'Output' section at the bottom shows the 'Action Output' table:

#	Time	Action	Message
56	15:42:36	SELECT * FROM orders WHERE salesman_id=5...	4 row(s) returned
57	15:43:30	SELECT * FROM orders ORDER BY purch_amt ...	12 row(s) returned
58	15:44:22	SELECT * FROM orders ORDER BY ord_date LI...	12 row(s) returned
59	15:45:02	SELECT SUM(purch_amt) AS total_purch_amt FRO	1 row(s) returned

8.LOWEST ORDER AMOUNT

The screenshot shows the MySQL Workbench interface with the following components:

- Navigator:**
 - MANAGEMENT:** Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, Data Import/Restore.
 - INSTANCE:** Startup / Shutdown, Server Logs, Options File.
 - PERFORMANCE:** Dashboard, Performance Reports, Performance Schema Setup.
- Administration / Schemas:** Information, Object Info, Session.
- SQL Editor:** Contains two queries:

```
145 • SELECT MAX(purch_amt) AS highest_order
146 FROM orders;

149 • SELECT MIN(purch_amt) AS lowest_order
150 FROM orders;
```
- Result Grid:** Displays the result of the second query:

lowest_order
65.26
- Output:** Shows a log of actions and their results:

#	Time	Action	Message
✓ 57	15:43:30	SELECT * FROM orders ORDER BY purch_amt ...	12 row(s) returned
✓ 58	15:44:22	SELECT * FROM orders ORDER BY ord_date LI...	12 row(s) returned
✓ 59	15:45:02	SELECT SUM(purch_amt) AS total_revenue FRO...	1 row(s) returned
✓ 60	15:45:41	SELECT AVG(purch_amt) AS average_order FRO...	1 row(s) returned
✓ 61	15:46:26	SELECT MAX(purch_amt) AS highest_order FRO...	1 row(s) returned
✓ 62	15:47:05	SELECT MIN(purch_amt) AS lowest_order FROM...	1 row(s) returned

9.TOTAL ORDERS

The screenshot shows the MySQL Workbench interface with the following components:

- Navigator:** Contains sections for MANAGEMENT (Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, Data Import/Restore), INSTANCE (Startup / Shutdown, Server Logs, Options File), and PERFORMANCE (Dashboard, Performance Reports, Performance Schema Setup).
- SQL Editor:** Displays three queries:

```
145 • SELECT MAX(purch_amt) AS highest_order
146 FROM orders;
147
148
149 • SELECT MIN(purch_amt) AS lowest_order
150 FROM orders;
151 • SELECT COUNT(*) AS total_orders
152 FROM orders;
153
```
- Result Grid:** Shows the results of the third query:

total_orders
12
- Output:** A table showing the execution of multiple queries:

#	Time	Action	Message	Duration /
✓ 58	15:44:22	SELECT * FROM orders ORDER BY ord_date LI...	12 row(s) returned	0.000 sec
✓ 59	15:45:02	SELECT SUM(purch_amt) AS total_revenue FRO...	1 row(s) returned	0.000 sec
✓ 60	15:45:41	SELECT AVG(purch_amt) AS average_order FRO...	1 row(s) returned	0.000 sec
✓ 61	15:46:26	SELECT MAX(purch_amt) AS highest_order FRO...	1 row(s) returned	0.000 sec
✓ 62	15:47:05	SELECT MIN(purch_amt) AS lowest_order FROM...	1 row(s) returned	0.000 sec
✓ 63	15:47:49	SELECT COUNT(*) AS total_orders FROM orders ...	1 row(s) returned	0.032 sec

10.TOTAL SALES GRANTED

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. Below this is the 'INSTANCE' section with Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes Dashboard, Performance Reports, and Performance Schema Setup. The bottom of the sidebar has 'Administration' and 'Schemas' tabs, and an 'Information' section that currently says 'No object selected'.

The main window displays a SQL query in 'SQL File 6'. The query is:

```
153
154
155
156 • SELECT salesman_id,
157       SUM(purch_amt) AS total_sales
158 FROM orders
159 GROUP BY salesman_id;
160
161
```

Below the query editor, the 'Result Grid' is shown with the following data:

salesman_id	total_sales
5002	1349.45
5001	11271.46
5003	2590.90
5005	270.65
5006	1983.43
5007	75.29

At the bottom, the 'Output' section shows a list of actions and their messages:

#	Time	Action	Message
59	15:45:02	SELECT SUM(purch_amt) AS total_revenue FRO...	1 row(s) returned
60	15:45:41	SELECT AVG(purch_amt) AS average_order FRO...	1 row(s) returned
61	15:46:26	SELECT MAX(purch_amt) AS highest_order FRO...	1 row(s) returned
62	15:47:05	SELECT MIN(purch_amt) AS lowest_order FROM...	1 row(s) returned
63	15:47:49	SELECT COUNT(*) AS total_orders FROM orders ...	1 row(s) returned
64	15:48:30	SELECT salesman_id, SUM(purch_amt) AS t...	6 row(s) returned

11.TOTAL PURCHASES MADE

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 5* SQL File 6* x SQLAddit

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information: No object selected

SQL File 6* x

```

159 GROUP BY salesman_id;
160
161
162 • SELECT customer_id,
163       SUM(purch_amt) AS total_purchase
164 FROM orders
165 GROUP BY customer_id;
166
167

```

Limit to 1000 rows

Result Grid

customer_id	total_purchase
3005	1099.00
3002	8870.86
3009	2590.90
3007	2400.60
3001	270.65
3004	1983.43
3003	75.29
3008	250.45

Result 32 x

Output

Action Output

#	Time	Action	Message
✓ 60	15:45:41	SELECT AVG(purch_amt) AS average_order FRO...	1 row(s) returned
✓ 61	15:46:26	SELECT MAX(purch_amt) AS highest_order FRO...	1 row(s) returned
✓ 62	15:47:05	SELECT MIN(purch_amt) AS lowest_order FROM...	1 row(s) returned
✓ 63	15:47:49	SELECT COUNT(*) AS total_orders FROM orders ...	1 row(s) returned
✓ 64	15:48:30	SELECT salesman_id, SUM(purch_amt) AS t...	6 row(s) returned
✓ 65	15:49:14	SELECT customer_id, SUM(purch_amt) AS t...	8 row(s) returned

Object Info Session

12.CUSTOMER MAX PRICES

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 5* SQL File 6* x SQLA

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information: No object selected

SQL File 6* x

Limit to 1000 rows

```
166
167
168 • SELECT customer_id,
169         MAX(purch_amt) AS max_purchase
170 FROM orders
171 GROUP BY customer_id
172 HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
173
174
```

Result Grid

	customer_id	max_purchase
▶	3002	5760.00
	3009	2480.40
	3007	2400.60

Export: Wrap Cell Content

Result Grid

Form Editor

Field Types

Result 33 x

Read Only

Output

Action Output

#	Time	Action	Message
✓ 61	15:46:26	SELECT MAX(purch_amt) AS highest_order FRO...	1 row(s) returned
✓ 62	15:47:05	SELECT MIN(purch_amt) AS lowest_order FROM...	1 row(s) returned
✓ 63	15:47:49	SELECT COUNT(*) AS total_orders FROM orders ...	1 row(s) returned
✓ 64	15:48:30	SELECT salesman_id, SUM(purch_amt) AS t...	6 row(s) returned
✓ 65	15:49:14	SELECT customer_id, SUM(purch_amt) AS t...	8 row(s) returned
✓ 66	15:50:17	SELECT customer_id, MAX(purch_amt) AS m...	3 row(s) returned

Object Info Session

